

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2017.Doi Number

A Survey on Cross-Chain Technology: Challenges, Development, and Prospect

HANYU MAO, TIEZHENG NIE, HAO SUN, DERONG SHEN, AND GE YU (Senior Member, IEEE)

School of Computer Science and Engineering, Northeastern University, Shenyang 110819, China

Corresponding author: Tiezheng Nie (e-mail: nietiezheng@mail.neu.edu.cn).

This work was supported in part by the National Natural Science Foundation of China under Grant 62072082, in part by the National Social Science Foundation of China under Grant 21&ZD124, in part by the Key Research and Development Program of Liaoning Province under Grant 2020JH2/10100037, in part by the Fundamental Research Funds for the Central Universities under Grant N2216008, and in part by the Fundamental Research Funds for the Central Universities under Grant N2216005.

ABSTRACT With the continuous development of blockchain technology, many blockchain applications, such as digital currencies in the form of tokens, are deployed. However, due to the lack of data and value inter-blockchain transmission methods, these chain applications form many data islands. Therefore, cross-chain technology is proposed and applied in distributed transaction platforms, digital finance, e-government and many other different fields. Although cross-chain technology just started, the cross-chain technology and its applications are gradually improving to meet the development needs of blockchain system. At the same time, they have already realized asset transfers across heterogeneous blockchains and platforms, and some projects have realized cross-chain information exchange. At present, cross-chain technology includes the following typical methods: Hash-locking, Notary scheme, Distributed private key control, Sidechains/Relays and Tight coupling. This paper analyzes the above methods from the aspect of techniques, security requirements, and technical limitation, and compares their applicability. With public blockchains and consortium blockchains, this paper analyzes their problems and challenges in cross-chain technology and introduces the main approaches of existing solutions. By analyzing the existing cross-chain technologies and problems, this paper further introduces eight popular cross-chain projects and their application prospects. Finally, this paper summarizes and proposes the future research topics and development trends of cross-chain technology.

INDEX TERMS Blockchain, Cross-chain Transaction, Information Transfer, Interoperability.

I. INTRODUCTION

Blockchain technology realizes decentralized shared general ledger which is difficult to tamper with and unforgeable by security mechanisms such as public key encryption, hash chain and proof of work consensus. A blockchain system uses digital signatures and encryption algorithms to generate blocks containing data and relies on Consensus to ensure that the entire participant nodes (may be the majority of nodes) recognize the blocks to ensure their safety and reliability [1, 2].

Blockchain can be understood as a distributed database used and maintained by all participants in the network. So that, it is traceable, verifiable and not easy to be tampered. However, different from the traditional database systems, all participants have obtained all the data on chain, which improves the query

efficiency of participants and the data security of the system and reduces the maintenance cost.

Public blockchains use decentralized approaches, such as Bitcoin blockchain nodes participating in transactions by the PoW [3,4] consensus, jointly maintaining the integrity of the public blockchain and the persistence of transactions, ensuring the credibility and traceability of data. Commonly used consensus strategies include POS [4,5], DPOS [6], Casper [7,8,9], etc. A consortium blockchain is controlled by a group of preselected nodes. The preselected nodes are highly credible, also control the permissions of the joining nodes. For example, in the PBFT [10, 11] consensus, the number of signature nodes reaches two-thirds of the total number will be accepted, and the information is not necessarily public, such as root hash value of public block. A consortium chain belongs to weakly centralized blockchain. Although reducing some

decentralization may have an impact on security [12], a consortium chain still reduces the verification burden and transaction cost to improve the handling capacity and providing more flexible blockchain structures. Commonly used consensus strategies include Paxos [13], RAFT [14], PoA (Proof of Authority) [15], Ripple [16], etc. In consortium chain, there are different permission levels to control the access permission of nodes, so that the consortium chain can provide a higher level of privacy protection [17].

Recently, blockchain develops fast, many blockchain projects with their unique chain structures and tokens have been created, which do not interoperate with each other and do not have any connection interfaces. This leads to a great obstacle to the application and development of blockchain technology. These tokens and on-chain application data may form blockchain data islands, so different blockchains need a unified method to transfer data and value between them. Researchers have proposed the concept of "Cross-chain" to break the data island between different blockchain projects and realize interoperability between chains. Blockchain technology has endless potential in real-time payment, financial transaction, Internet of Things, etc., and the Smart Contract based on Ethereum and the Chaincode of Hyperledger Fabric [18,19] have become the core of technology implementation. In the regulatory field, the traceability of blockchain can be further extended with the support of cross-chain technology [20]. The current cross-chain transactions overly rely on the middleman mechanism, which weakens the decentralization. Cross-chain mechanisms such as Relays can send the cross-chain transaction data to each node according to the underlying blockchain technology, making the whole transaction process clear and transparent. Especially in consortium blockchains, transactions using Distributed Private Key Control or Relays are supervised by consortium nodes, which makes the transaction process credible and legitimate. Through the collaboration of member nodes and regulatory agencies, it can ensure efficient and transparent regulatory information processing and improve the management ability.

Cross-chain technology can be understood as a chain-to-chain communication protocol. Current cross-chain technologies include the following approaches: Notary schemes, Sidechains/Relays, Distributed Private Key Control, Hash-locking, and Tight Coupling. In this paper, we survey the technical principles of each cross-chain technology, make analysis and comparison with them. Some related research works have summarized the blockchain cross-chain technology, He [21] and Lu [22] introduced the basic cross-chain technology, Xu [23] introduced the cross-chain classic project, Guo [24] analyzed the existing cross-chain technology and project, Li [25] analyzed the cross-chain technology projects from the security perspective. Lin [26] introduced the cross-chain technology architecture and cross-chain technology. Of their work, we make a further improvement in the systematicity and completeness of the latest cross-chain

technologies and related applications, and compare and analyze different technologies and applications in terms of performance, security, decentralization degree, etc.

This paper attempts to summarize the five state-of-the-art cross-chain technologies and compare them from multiple perspectives systematically and comprehensively. In addition to analyzing the challenges of cross-chain technologies, we introduce the key underlying technologies involved, including consensus mechanisms and security issues, and introduce the popular and emerging cross-chain projects, while comparing and evaluating the performance of these projects.

Section 1 introduces and identifies the similarities and differences between public blockchain and consortium blockchain and explains the concept of cross-chain technology. Section 2 introduces the current research focus of cross-chain technology and discusses the challenges of current cross-chain technology research. Section 3 discusses the five cross-chain technologies, Hash-locking, Notary Schemes, Distributed Private Key Control, Sidechains/Relays and Tight Coupling, finally analyses and compares them. Section 4 introduces the mainstream projects and applications of cross-chain technology, introducing the current research and development of cross-chain technology and the mainstream research projects, as well as what improvements and innovations these projects have made in the kernel algorithms, making a side-by-side comparison of the cross-chain projects and introducing related emerging applications. Section 5 introduces the development trend of cross-chain technology and proposes the future development direction and research hotspots of cross-chain technology. Section 6 concludes paper.

II. CHALLENGES OF CROSS-CHAIN TECHNOLOGY IN BLOCKCHAIN SYSTEMS

The current blockchain technology is developing rapidly, the existing mature blockchain systems, such as Ethereum, Fabric, Quorum, etc., realize some of the blockchain cross-chain functions, but still have more limitations and problems to be solved. The following are the challenges of cross-chain technology in blockchain systems in current applications.

A. CROSS-CHAIN INTERACTION PROTOCOL

With the development of cross-chain technology of blockchain, users need to obtain information from other blockchain systems, but a single blockchain system is a closed system, the authorization and verification of information is only valid within the system, information cannot be obtained from outside, and the outside cannot obtain any information inside the single chain, which is the "data island" effect. Cross-chain technology can solve the "data island" effects generated by single chain, since cross-chain technology is essentially to send the data X on chain A to chain B securely and trustworthy, and can query, store and update information on B. From the data storage layer, the on-chain asset (token) is a piece of encrypted and tamper-resistant data. Under the traditional centralized ledger structure, such data is modified and queried

by the administrator and the system, but considering the blockchain structure, the cross-chain data and information interaction requires precise bookkeeping operations, so a reasonable cross-chain protocol is needed to ensure the atomicity, consistency and data security of operations in each blockchain system.

B. TRANSACTION PROCESSING PERFORMANCE OF PUBLIC CHAIN

The performance bottleneck problem of public chain transaction processing is a classical problem in blockchain technology, such as the block issuance interval of Bitcoin blockchain is about 10 minutes [27], which is determined by the mining difficulty factor determined by its PoW consensus, and the corresponding difficulty factor will increase if the computing power of the whole chain is enhanced. The Ethereum chain has decreased the block-out interval to about 15 seconds, but it is still unbearable for the huge transaction performance requirements of current blockchain applications. The block capacity demand of blockchain has led to a large number of blockchain extension schemes [28,29], but researchers believe that simply increasing the block capacity will lead to an increase in the operating cost of blockchain systems, and the improvement in throughput performance is not obvious [30,31].

The performance problem of transaction processing in public chain systems has seriously restricted the applications of blockchain technology, for this reason, scholars have proposed Sharding, DAG, Off-Chain Computing, Sidechains and other methods. For example, Lightning Network uses Off-Chain Computing. Lightning Network intermediate transactions are not confirmed through the mainchain and the results are packaged on chain. This method can observably decrease the delay time of small-scale on-chain transactions. The purpose of Sidechain is to extend the functionality and performance of the mainchain. Sidechain was first proposed to achieve joint mining of public blockchains, later developed to improve the performance of the mainchain. For example, using Plasma technology to build sidechain on Ethereum. Plasma is designed to be similar to a kind of network sharding, or consensus, booking, and computational processing in sharding. This method can enforce state locking by writing smart contracts on the mainchain on the mainchain, using proof of fraud. By forcing a chain's account information to be packaged into a sub-blockchain through the mainchain, the chain will be incredibly scalable with minimal trust.

However, the performance improvement of cross-chain technology is still limited by the mainchain consensus, block size and other issues, which are the main research challenges at present.

C. EXTENSIBILITY OF CONSORTIUM CHAIN

Consortium chains can decide the degree of openness to the public according to the application scenario, and its network is jointly maintained by the member institutions, and the nodes

are accessed through the gateway nodes of the member institutions. So it is suitable for the direction of storing, managing, authorizing, monitoring and auditing dynamic data by multiple member institutions. In the context of practical applications, users, resources, services and terminals are characterized by access and authorization operations [32]. Due to the above-mentioned characteristics, there are huge differences in the design of consensus, authority control, chain code design and node control of the coalition chain, which has long restricted the development of blockchain technology. Therefore, the consortium chains need to be developed through a unified cross-chain protocol as an industry paradigm and blockchain-as-a-service architecture platform (BaaS) [33] to solve the scalability problem.

The current consortium chain technology is widely used in finance, government, healthcare, education and other scenarios, and developers need to continuously add or change its functions to fit the application scenarios. The traditional way is to add a new Chaincode to a consortium chain [34,35], but this method increases the workload of a single chain and reduces the operation rate and concurrency efficiency of the chain. A common method to extend the functionality is to implement or cross-chain operation through Sidechains or multi-chain architecture, thus avoiding the above problems and improving the security of the main chain while ensuring the operability and efficiency of the main chain. Therefore, the cross-chain technology needs to further solve the parallelism of multi-chain architecture to improve the scalability of the federated chain under the condition of ensuring security.

D. CROSS-CHAIN BOOKKEEPING AND CLEARING

Cross-chain technology can use distributed mechanisms to solve the problem of clearing between multiple chains, and cross-chain transaction clearing means calculating and verifying the assets on the chain. The original design of Bitcoin is that the whole blockchain system uses a common set of tamper-proof distributed ledger, however, different blockchain systems have their own set of ledger and bookkeeping clearing mode, and decentralized transactions should ensure that funds do not need to be escrowed, users do not need to register. This poses a challenge to cross-chain token exchange. Good cross-chain technology can link different blockchain systems together, so that different systems can still access each other and even keep the ledgers of the counterparties and avoid the control of the exchange as a centralized node for different on-chain systems, to achieve a more decentralized transaction process.

Cross-chain technology-based bookkeeping and clearing can realize blockchain distributed transactions, enabling the merging of real financial markets with digital financial assets and reducing additional transaction costs when passing through centralized exchanges. Currently, in the application of bookkeeping and transaction clearing, cross-chain technology faces the challenge of how to achieve a unified and efficient bookkeeping and clearing function in a multi-chain

environment and ensure the decentralized nature of the multi-blockchain ecosystem.

III. ANALYSIS OF CURRENT CROSS-CHAIN RESEARCH STATUS

Cross-chain technology can be divided into two classes: heterogeneous cross-chain and homogeneous cross-chain. Heterogeneous cross-chain can be divided into three ways: no direct interaction, third-party collaborative interaction and block listening. In homogeneous cross-chain, the underlying algorithms such as block generation, block verification and consensus are identical between the two parties. This paper focuses on the cross-chain technology of heterogeneous blockchains.

No direct interaction cross-chain is shown in Figure 1.a. The representative method is Hash-locking, and the typical application is Lightning Network, which is based on micro-payment channel [36] to achieve asset transfer, where a smart contract is used to control cross-chain asset exchange without involving both blockchain structures.

The third-party collaborative interaction cross-chain is shown in Figure 1.b, which specifically includes Notary Authentication and Distributed Private Key Control, through single to multiple third-party nodes involved in cross-chain interaction, are applicable to the scenario of mutual distrust participants. The architecture mostly relies on the centralized nodes of third parties to achieve.

The block listening cross-chain is shown in Figure 1.c, which includes sidechain/relay technology and tight coupling mode, among which listening to the target chain by saving the target block header to achieve cross-chain operation. Which

requires anchoring the target chain data, which is more difficult to implement.

A. HASH-LOCKING

Hash-locking [37] uses hash algorithm [38] and time lock [39] to achieve cross-chain asset exchange, which is often applied to exchange tokens on different chains, such as between ethereum and bitcoin on ethereum, but the needs of asset transfer and message sending cannot be achieved independently. Hash-locking provides time delay for both parties to interact assets and ensures the atomicity of transactions [40], i.e., the operation of both parties is a whole and can only be fully completed or fully undone, and the priority encryption of sent assets avoids problems such as double-spending, which is a lightweight and more secure cross-chain asset interaction technology.

1) PRINCIPLE OF HASH-LOCKING TECHNOLOGY

The hash-locking uses hash lock and time lock technology to implement the currency exchange function, both parties need to lock the relevant tokens, the receiver will determine the receipt within the time limit and send the proof of receipt to the initiator, the initiator uses the proof of receipt to unlock and obtain the locked tokens from the receiver.

Take the example of bitcoin and eth exchange: at a certain moment 1 bitcoin can be exchanged for 27.85 ethereum, at this rate if the user Barry needs to use 1 bitcoin to exchange 27.85 ethereum of the user Yuri, the hash lock execution process [41,42] is shown in Figure 2, the specific steps are as follows:

a. Barry generates a random number m (password) for hashing, generates a hash value M , and sets a waiting time $t1$;

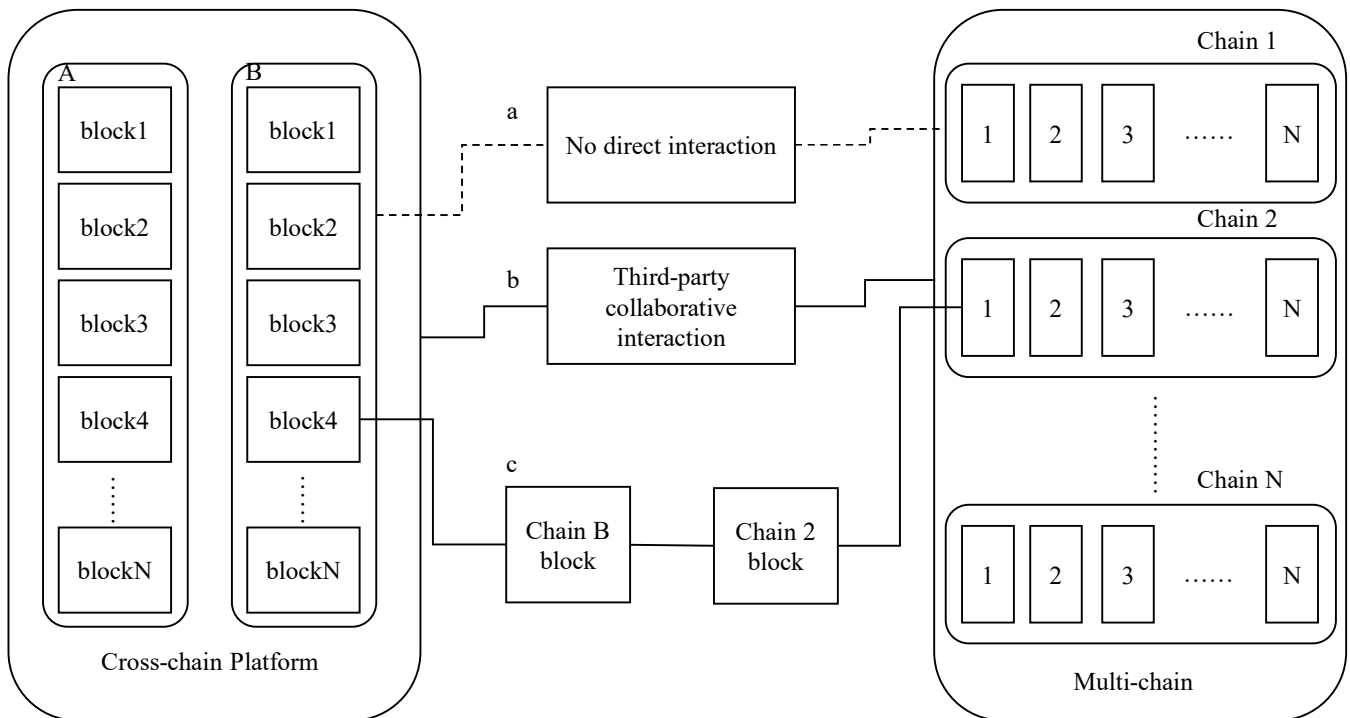


FIGURE 1. Heterogeneous cross-chain

- b. Barry uses M and wait time t_1 to generate a contract C_1 on Barry's chain A. C_1 locks Barry's 1bitcoin and waits for time t_1 ;
- c. Barry sends the hash value M to Yuri, and at the same time sends the transaction contract C_1 , broadcasting that he has locked 1bitcoin, with a waiting time of t_1 , beyond which the transaction fails and is revoked;
- d. Yuri sets another waiting time t_2 , which is less than t_1 , and uses M and t_2 to generate contract C_2 on Yuri's chain B. C_2 locks 27.85eth and waits for t_2 ;
- e. Barry uses the random number password m to unlock the transaction contract C_2 and its assets on chain B and generate contract C_3 , C_3 is used to send the password m that unlocks the assets on B to Yuri, Barry gets the locked 27.85eth on chain B, C_3 sends the password m to Yuri;
- f. Yuri uses the password m to unlock the transaction contract C_1 and its assets on chain A (which requires the creation of contract C_4), and obtains the locked 1bitcoin on chain A;

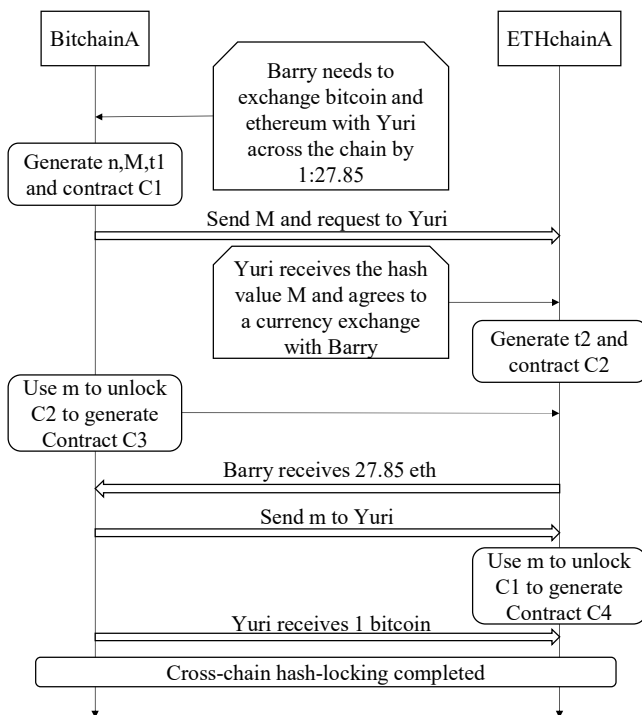


FIGURE 2. Hash-locking processing

Hash-locking sets a time lock to prevent delays in situations such as network congestion or node failure, and t_2 needs to be less than t_1 in the exchange method step to prevent the requesting party A chain (Barry) timeout to cancel the transaction after B chain is still waiting for the situation. Hash-locking is more lightweight and secure and used in many blockchain wallets.

2) SECURITY REQUIREMENTS OF HASH-LOCKING

The security requirements of Hash-locking [43] mainly include the following aspects: the participants are required to verify the security of the ledger information, and the traceability of the blockchain avoids the possibility of nodes

being cheated by forged blockheads in SPV mode. Hash-locking technology has good transaction atomicity to avoid double-spending problem. Time limit is set up to monitor the time on the chain in real time to prevent network congestion or invalid nodes from blocking subsequent transactions. Verifying the identity of cross-chain participants is to prevent malicious nodes from disguising and conducting spoofing attacks.

3) HASH-LOCKING TECHNOLOGY LIMITATIONS

Hash-locking technology has the following limitations [43]: the application scenario is single, currently can only be applied to cross-chain asset exchange. In the transfer of assets, information transfer, cross-chain contracts and other applications have not been realized. Essentially Hash-locking ensures that the sum of assets on the two chains remains unchanged, can not actually transfer assets, information, etc., need a third-party notary and other methods to cooperate with the realization. Due to the time delay, a participant can choose whether to complete the transaction according to the exchange rate change, so there is a fairness problem for both sides of the transaction. The implementation cost is too high, which of a transaction (such as the lightning network) is equivalent to the need for two cross-chain transactions and four smart contracts on the chain, which may lead to high transaction fees.

Hash-locking technology [44] is often used in conjunction with micro-payment channel technology (stateful channels), such as the Lightning Network, to collect cryptographic signature information by transferring transactions on the blockchain and interacting with participants in the corresponding payment channel. When the payment channel is closed, the final state will be on the chain, while the intermediate state will not be published in the chain of transactions between the two parties, and the problem of chain disputes may arise when it is closed [45].

B. NOTARY SCHEMES

Notary authentication [46] assumes that the sending chain A and the target chain B cannot trust each other, and a third-party notary is required to intervene as an intermediary in the transaction between them. If the system has higher security requirements, more third-party notaries are required to intervene. The target chain B does not need to directly verify the secret key information of the sending chain A to confirm whether the cross-chain request is legitimate, this method decouples the two chains, and the two chains only need to set up a unified protocol notary certification contract interface to meet the third-party notary intervention conditions. At the application level, the notary is a centralized exchange [47] or a decentralized exchange [48], and a single notary can monitor multiple chains. At the technical level, notary authentication technology is relatively simple to implement, and notary technology uses a two-way cross-chain approach to reduce the cost of trust between two parties for the exchange and transfer of assets and information, but the notary authentication mechanism is overly dependent on third-party centers, which

weakens the decentralized nature of blockchain technology. Notary schemes can be divided into centralized notary, multi-signature notary and distributed signature notary.

1) CENTRALIZED NOTARY

The centralized notary method [44], which is also known as the single-signature notary, selects only one third-party node as the third-party notary, which is usually a designated organization, and the user sends the funds and information to the third-party notary, who realizes the cross-chain operation between A and B. This method is limited by the third-party centralized node, which ignores the decentralized nature of blockchain and affects the overall security of the blockchain system.

2) MULTI-SIGNATURE NOTARY

This method uses multiple nodes as notaries [47], and uses a multi-signature mechanism, so that cross-chain operations can be performed only after the selected notaries reach a consensus and sign their respective ledgers. In the multi-signature mechanism, a ratio can be set, i.e., the cross-chain transaction can be carried out once the number of selected notaries reaches this ratio, and each notary has a different private key to ensure the security of the system. Compared with centralized notaries, the multiple signature mechanism improves the overall security of the system and preserves the decentralized nature of the blockchain, although it sacrifices some of the transaction speed.

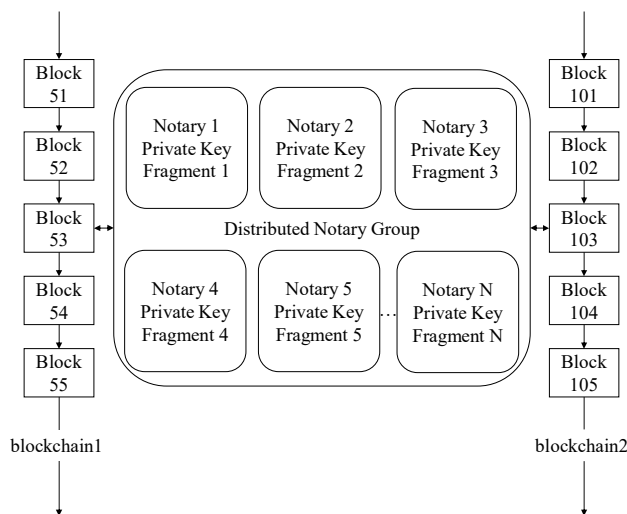


FIGURE 3. Distributed Signature Notary Technology Implementation Architecture

3) DISTRIBUTED SIGNATURE NOTARY

The distributed signature notary [49] also selects multiple nodes as the notary, the difference is that each notary only has a random private key fragment sent by the system, the private key fragment has been encrypted, even if multiple notary nodes are not trusted, it is impossible to obtain a valid complete private key, further decentralization and improve the security of the system. Figure 3 shows the implementation architecture of distributed signature notary technology, the system is based on cryptography students into a secret key

with one and only one, and split it into N copies, distributed to a randomly selected group of notaries to form a distributed notary, and then by their co-signatures in their respective ledgers to reach consensus, co-signatures need to reach a certain number or proportion to complete cross-chain transactions.

4) SECURITY REQUIREMENTS OF NOTARY AUTHENTICATION MECHANISM

Firstly, the scheme needs to ensure the security and trustworthiness of the notary nodes to avoid the "evil" of the notary, and secondly, the security of both chains A and B is checked, i.e., the latest information of the block issuer collection provided by the public chain system and the member node information provided by the consortium chain system are checked [44]. Before the cross-chain contract goes online, the notary access license similar to the consortium chain is implemented, and the centralized notary approach is avoided as much as possible to ensure the decentralized nature of cross-chain transactions. The number of notaries should be more than the number of sending chain A consensus nodes to ensure the security of the notary environment in cross-chain transactions, and to establish an incentive mechanism for notary nodes, such as the faster and more accurate the secret key verification consensus signature can be rewarded with tokens, while the nodes with poor computing power that continue to sign multiple transactions in a timely manner may be replaced by the high computing power security nodes waiting for access [50].

5) NOTARY SCHEME LIMITATIONS

Users need to have enough trust in the notary scheme to use it, and there are risks of fund and information leakage. There are significant centralization risks in using centralized notary scheme, similar to using a third-party exchange [44]. The multi-signature notary scheme needs to be supported on the target chain, and there are technical barriers for some blockchains.

C. DISTRIBUTED PRIVATE KEY CONTROL

Distributed private key control technology uses a large number of distributed nodes and uses private keys to control the related assets, and the main chain assets can be mapped to other chains, and the main chain assets can still be traded in the cross-chain scenario. Distributed private key technology further decentralizes than the distributed signature notary and adds the function of locking and unlocking assets. Distributed private key technology uses decentralized network nodes to achieve distributed control management, which separates the ownership and usage of assets on the chain [51]. Each participant of the decentralized network holds a part of the user's private key to ensure the security and reliability of the system, and the user also holds a part of the decentralized network's private key, thus avoiding the participation of third-party nodes in the control and realizing the decentralized nature of the system.

1) DISTRIBUTED PRIVATE KEY CONTROL SCHEME

Distributed private key [52] checks the transferred-in and transferred-out amounts during cross-chain transactions, and the system immediately withdraws the funds and cancels the transactions by means of private key collection when there is a problem with the amounts to ensure the atomicity of the transactions. In terms of distributed technology, distributed private key technology is similar to but better than distributed signature notary, which uses a decentralized distributed network to save the private key in pieces, realizing the requirement of decentralization of blockchain technology, even if the on-chain network is partially attacked, the attacker cannot use the partial secret key to obtain the complete user secret key. In the distributed private key control, the user can unlock the transaction by using K secret key slices in the system (the number of K is defined by the developer, less than or equal to the total number of slices), which ensures the fast processing of transactions and solves the problem of malicious node lag, and if a node fails to complete the private key verification for a long time, it can be replaced by the reward and punishment mechanism for the node located in the distributed network.

2) SECURITY REQUIREMENTS OF DISTRIBUTED PRIVATE KEY CONTROL

Distributed private key technology uses distributed secret key generators [53], and its decentralization is better than distributed signature notary, so the security requirements for third-party nodes are lower than distributed signature notary. To prevent double-spending problem [54], it needs to wait for the finalization of on-chain transactions before starting verification, sacrificing some efficiency to ensure asset security.

3) DISTRIBUTED PRIVATE KEY CONTROL LIMITATIONS

Since the characteristics of the original chain are not changed, the cross-chain development needs to adapt to the characteristics of the original chain, so the development is difficult, and the waiting time for the confirmation of the original chain is long, resulting in low operation efficiency [53]. The distributed private key control is similar to the notary scheme, but the user always has the control over the assets. Distributed storage [51] is adopted to store the key to digital assets, which avoids the risk of centralization under notary mechanism to a certain extent. But again, this makes development much more difficult.

D. SIDECHAINS/RELAYS

Sidechain is a scheme to interoperate [55,56], extend [57] or upgrade [58] two existing blockchains, usually one blockchain (sidechain) anchors another target blockchain (main chain), and the sidechain is attached to the main chain as an independent system through a cross-chain protocol [59]. The relays can be regarded as a sidechain anchored to multiple master chains, providing token relay exchange services for multiple chains.

1) SIDECHAINS

Sidechain as a parallel chain to the main chain helps the main chain to achieve performance improvement and function expansion [60], which includes some simple payment verification to improve the transaction speed of the main chain. The most widely used blockchain system that applies sidechain technology is Ethereum, such as Bitcoin blockchain as main chain A and Ethereum blockchain as sidechain B. B can read A, while main chain A does not know the existence of B. Sidechain can be understood as a kind of extension protocol based on the main chain, mainly used to extend the digital currency blockchain. Nowadays, the "Two-way Peg" technology [61] has been developed, based on which the anchored sidechain can circulate assets and information in both directions between chain A and chain B. The most widely used one is the SPV proof mode anchoring, which will be introduced in subsection 3.4.3. In the technical report of R3 corda [62], it is proposed that two master chains can be used as sidechains for each other to achieve the function of Two-way Peg [63].

2) RELAYS

When the sending chain A and the target chain B cannot communicate directly, and do not want to use the notary, they can use at least one relay chain that replaces the intermediary authentication for communication, and if a sidechain links multiple master chains and can provide the function of communicating each master chain, the chain can be called a relay chain [64]. Because the relay chain technology is implemented on the chain, it has the characteristics of flexibility and high scalability, and has realized cross-chain contracts, asset collateral, cross-chain asset transfer, cross-chain information interaction, etc. Currently, relays are widely used in cross-chain applications, and Cosmos and Polkadot are common and more mature relay chain technologies.

3) SPV

Simplified Payment Verification [65,66,67] (SPV), the SPV only needs to confirm whether the transaction has been verified and how many digital signatures the transaction has been confirmed. Therefore, SPV is different from notary, which only needs to verify how many nodes on the sending chain A have confirmed the payment, and this method greatly improves the efficiency of transaction processing in the sidechain. The SPV verification process can be divided into the following five steps, as shown in Figure 4.

- a. The SPV node calculates the received waiting transaction hash value for verification of payment;
- b. The SPV node obtains the Blockheader of the current longest chain of the blockchain network to local;
- c. The SPV node obtains the Hash authentication path of the MerkleTree [68] to be verified;
- d. Check if the blockheader is on the longest chain;
- e. Based on this blockheader, get the number of digital signature confirmations (at least 6 block confirmations) for the payment to be verified;

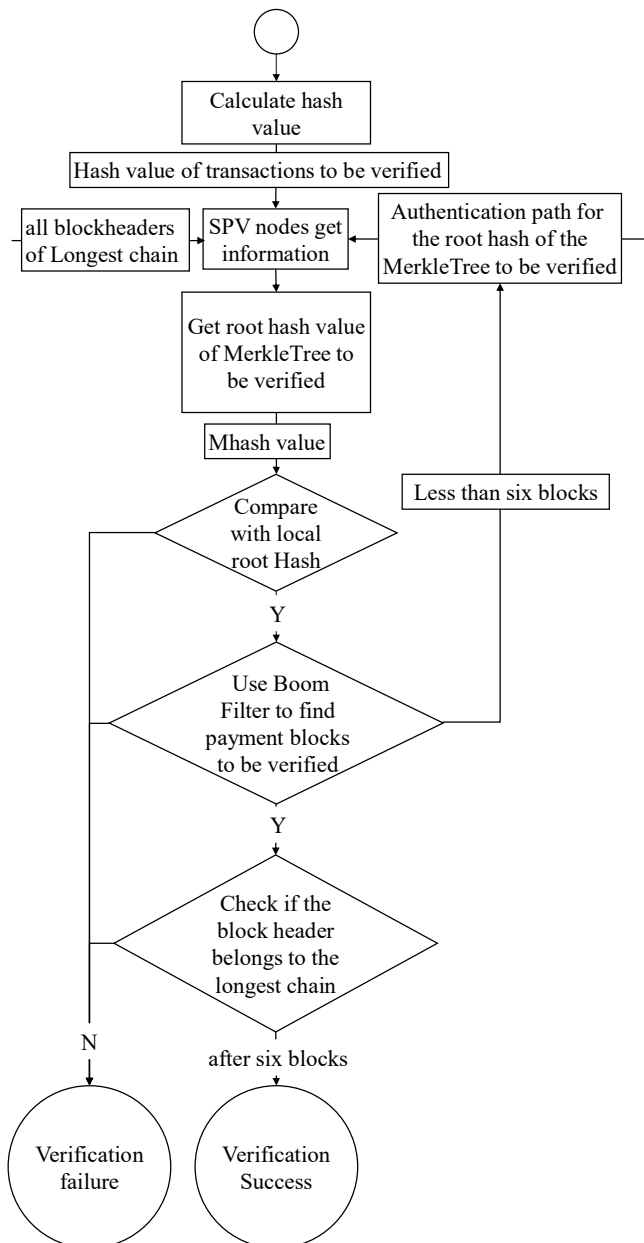


FIGURE 4. SPV node validation process

Applying the SPV to sidechains can be called SPV Two-way Peg technology [69], and its verification process can be divided into the following steps, as shown in Figure 5.

- The main chain creates the transaction and locks the associated asset (token) on the main chain;
- Wait long enough to ensure that the transaction receives sufficient proof of workload and digital signatures, such as the six block confirmations mentioned above, to resist double-spending attacks;
- When the waiting period in b is over, the sidechain generates a transaction, redeems the sidechain token according to the relevant ratio, sets the token generated by the redemption to a locked state, and waits for the end of the competition period;

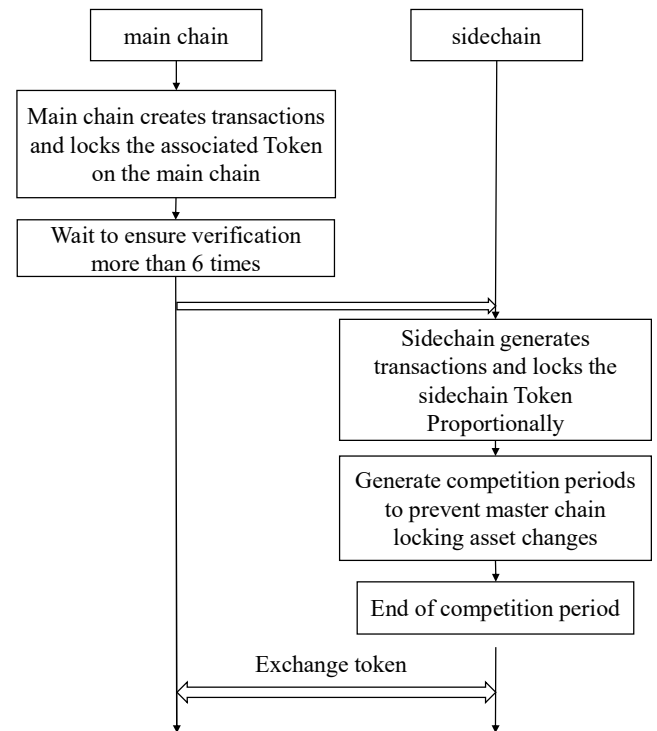


FIGURE 5. SPV Two-way Peg

d. The sidechain generates a competition period, in order to prevent the double-spending problem [70,71], if the locked assets of the main chain change during the competition period, the SPV node will receive the relevant confirmation information, and then the sidechain transaction fails and is revoked;

e. After the competition period, the sidechain token can be used for on-chain circulation;

f. After circulation, the sidechain token can be sent back to the main chain (after repeating steps a-e);

As in the BTC Relay system, participants send block header information from the main chain to a smart contract stored in an Ethereum container to create a pool of bitcoin chain block headers stored in Ethereum, which can be used to validate information on the chain. Each participant can request the smart contract holding the pool of block header information to validate the transaction in SPV mode, which can be used for token issuance and exchange between different tokens.

4) SECURITY REQUIREMENTS OF SIDECHAINS/RELAYS

When the Sidechains/Relays use SPV [65,67] for cross-chain transaction authentication, because SPV only saves block header data, which is very easy to be deceived by a forged block by a malicious node, and it is impossible to verify the existence of the transaction, so this vulnerability can be easily exploited by a double-spending attack. SPV mode [69] requires randomly finding and linking multiple blockchain nodes, which makes these nodes and transactions more vulnerable to network partitioning attacks.

5) SIDECHAINS/RELAYS LIMITATIONS

Sidechains/Relays [60,72] lead to additional complexity: many separate, asynchronous blockchains are added to the blockchain system at the network layer. At the asset level, a single chain can support multiple assets, and the chain needs to mark the source chain of each asset to ensure that the asset can be transferred correctly. Sidechain/Relays with miner fees may bring pressure to miner resources and lead to the risk of mining centralization. In addition, unless one sidechain is generally accepted, the resulting soft fork of the sidechain may lead to a failure of consensus.

E. TIGHT COUPLING

1) PRINCIPLE OF TIGHT COUPLING

The tight coupling mode [23] is to design the communication mode with other chains in the process of chain construction, to achieve a high degree of similarity or even unity in the design of contracts, interfaces, chain structure, node design, and block data structure, and to keep the block header information of other chains that need communication inside the chain before it goes online, so that only the internal block header information needs to be found when communicating across chains. This method is mainly used in internal networks, such

as military blockchain systems to prevent external access or attack, or private chain systems within the industry to prevent leakage of important data. And mining federated model is the main chain and the sidechain keep block header information with each other [23], and in the mining process, the sidechain works with the main chain to improve the arithmetic power of mining nodes, and the sidechain completes lightweight tasks, etc.

2) LIMITATIONS AND SECURITY REQUIREMENTS OF TIGHT COUPLING

Tight Coupling includes the block header information of the communicable chain at the time of design, although the process is easy to use, but its scalability is severely limited, smart contracts can not be better to achieve functional expansion, and cross-chain access to other chains that do not save the block header data may be subject to many restrictions; internal network using tight coupling mode may be more secure, and get a higher block issuance rate and response speed, etc., but the lack of good consensus mechanism and security control are needed to ensure the security of the internal network to ensure the security of information and assets of the whole system.

TABLE 1. Cross-chain Technology Performance Comparison

Cross-chain technology	Fee	Transaction Speed	Interoperability	Implementation Difficulty	Multi-token Contracts	Security	Interaction Information	Asset Transfer	Decentralization	Atomicity Requirements
Hash-Locking	High	Medium	Cross-dependency	Low	No support	High	No support	No support	High	HL timeout control
Notary scheme	Low(C) Med(M) High(D)	Low	Bidirectional	Medium	Difficult	Low(C) Med(M) High(D)	Support	Support	Low(C) Med(M) High(D)	Mutual Trust
Distributed Private Key	Low	Medium	Bidirectional	Medium	Support	Medium	Support	Support	High	Multi-signature
Sidechains/Relays	Low	Low	One-way (S) Bi (R)	High	Difficult	Medium	Support	Support	Medium	Merkle proof
Tight Coupling	Very low	High	One-way	Low	No support	Medium	Support	Support	High	No

F. PERFORMANCE COMPARISON

The above five main cross-chain application technologies all have different advantages and disadvantages. Table 1 compares the various aspects of performance of the five cross-chain technologies. Hash-locking only supports cross-chain asset transaction, does not support interaction information or asset transfer, but its implementation difficulty is very low, and the transaction is light and safe. Notary scheme can support multi-token and interaction information, and it is not difficult to achieve. But problems such as low speed, high commission, and obvious third-party centralization still need to be improved. Moreover, the notary scheme may not ensure the credibility of the third-party notary, of which the Centralized notary(C), Multi-signature notary(M), Distributed signature notary(D). Distributed private key technology supports multi-token and interaction information, which is easy to implement. However, it sacrifices the speed to wait for the verification on the blockchain to ensure the security of transaction and limited by the performance bottleneck of the block chain. Sidechain/Relays supports multi-token and

interaction information, which is difficult to implement. As the first generation cross-chain technology, the tight coupling mode is relatively simple to implement, but the performance guarantees such as scalability and interoperability are not satisfied. It is suitable for closed internal networks and improves mining computing power. The transaction fees in Table 1 are derived from actual tests. The transaction speed is determined by its design method, smart contract and on-chain exchange. From the interoperability, hash-locking belongs to cross-dependence, and both sides of the blockchain with atomic exchange need to use the same hash algorithm and HTLC protocol. However, in practical application scenarios, only a few blockchains meet the conditions. The notary scheme and distributed private key control support bidirectional operation through a third-party centralized organization, but this scheme greatly limits the decentralized nature of blockchain technology. Sidechain/Relays interoperability is determined by chain design. The tight coupling is designed to include the blockheader information of the communicable chain, so only one-way operation of the target chain is guaranteed. Based on the analysis of

implementation difficulty, hash-locking realizes cross-chain asset transfer through hash algorithm, time lock and smart contract, which is relatively simple to implement. Notary scheme needs to rely on the third-party notary as the intermediary to intervene in the transaction, and distributed private key control needs to use a large number of distributed nodes to realize the control management, so the implementation of the above scheme is moderately difficult. The essence of Sidechain/Relays is to target the emerging blockchain generated by the main chain, aiming to achieve performance improvement and function expansion. This scheme involves a variety of technologies such as task division and inter-chain parallelism, which is difficult to implement. The tight coupling mode contains the blockheader information of the target communication chain and the chain structure is similar. It is mostly used in joint mining and other scenarios, so the implementation difficulty is low. Multi-token contracts, interaction information, and asset transfer are determined by its smart contract algorithms and code. The degree of security and decentralization depends on aspects such as reliance on third parties and contract locking. The atomicity requirement is a central feature of the above techniques.

IV. MAINSTREAM PROJECTS AND APPLICATIONS OF CROSS-CHAIN

The current mainstream projects of cross-chain technology include Cosmos, Polkadot, Interledger, etc. The purpose is to realize asset exchange, information interaction and application collaboration between different blockchain platforms, reduce the difficulty of blockchain users and reduce the cross-chain transmission process, and the current mainstream projects are as follows:

A. COSMOS

The Cosmos [73] project is developed by the Tendermint team based on public chains, using cross-chain token transfer technology. Cosmos chain is built from the Tendermint [74,75] consensus algorithm, which is an improvement of the Byzantine fault-tolerance algorithm that tolerates one-third of malicious or Byzantine nodes. The advantage of the Cosmos project is that it provides a blockchain network and the Tendermint BFT consensus algorithm package, which allows program developers to call directly and develop cross-chain programs without considering the underlying protocols, saving development time.

Cosmos SDK framework provides a large number of modules needed for development, such as the important Hub module in Cosmos. The Tendermint BFT package provided by Cosmos as a common engine for development also provides a socket protocol named ABCI (Application Blockchain Interface), which is used in conjunction with the Cosmos SDK development framework to provide a development interface for different languages, further improving the efficiency of program development.

The performance of the Cosmos is excellent, with a block generation time of about 1 second and the ability to process thousands of cross-chain transactions per second; in addition to being able to tolerate up to one-third of malicious or Byzantine nodes, the Tendermint consensus algorithm in Cosmos provides an accountability mechanism that identifies the responsible node and provides a solution when a fork occurs on the chain. The BFT engine allows developers to define on-chain verifiers, so when developing public, private and consortium chains based on the Cosmos project, developers are free to select verifiers and control permissions based on application requirements. In public chain projects, you can also select validators based on the number of tokens held on the chain, i.e., implement a POS proof-of-interest mechanism.

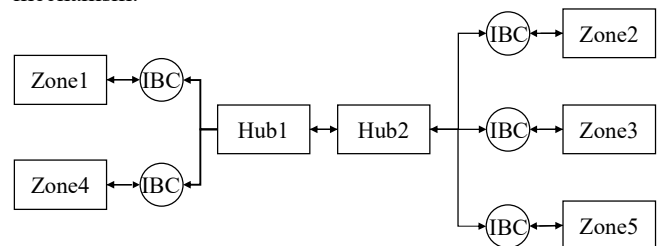


FIGURE 6. Cosmos blockchain network architecture

In cross-chain scenarios, heterogeneous chains are one of the major difficulties in cross-chain technology development, and developers need to implement relay chains in a "one chain, one relay" model. IBC [76] (Inter-Blockchain Communication Protocol) provided by Cosmos allows inter-chain communication and transfer of data and token between heterogeneous chains, thus achieving interoperability of heterogeneous chains. Cosmos also provides Hub chains, which are essentially trusted, efficient and extensible relay chains, whose purpose is to keep track of the status of each heterogeneous chain, and the heterogeneous chains need to send new blocks and ledger information from their own chains to the Hub. In Cosmos, heterogeneous chains are called Zone [76] application chains, and Zone chains need to connect to Hub chains through IBC protocol, and the Cosmos blockchain network is shown in Figure 6. The Hub broadcasts information about the newly added chain to other Zone chains in the Cosmos network after successful connection, and the Hub is very powerful, efficient and can prevent the double-spending problem, which realizes the trustworthiness of the Cosmos network.

B. POLKADOT

Polkadot [77,78], is developed by the ParityTech team, based on the relay chain approach, aiming achieving free communication between chains to enable the coexistence of multiple chains. Polkadot does not provide any functional extensions, but provides relay chains for consensus and data interaction, parallel chains for parallelizing the execution of on-chain applications, and bridge chains [79] to connect heterogeneous blockchains.

Polkadot consensus mechanisms include based on NPoS (Nominated Proof of Stake) [80], BABE (Blind Assignment for Blockchain Extension), and GRANDPA. NPoS consensus is used to elect an efficient and secure set of verifiers. BABE consensus is similar to the PoS consensus algorithm and acts as a block-out engine. GRANDPA consensus is used to complete block validation, which is characterized by multiple block validation at a time. BABE algorithm is based on slots, each of which is 6 seconds long. BABE selects a leader in each slot to block. When there is no leader, the leader is determined in a predetermined order. When there are multiple leaders, multiple nodes are allowed to commit blocks, with the final block acknowledgment at the discretion of GRANDPA, and Polkadot is shown in Figure 7. Polkadot mainly implements cross-chain communication by setting up channels between parallel chains and forwarding transactions in this way, which works as follows: When a parallel chain executes a transaction, it sends a second transaction to another parallel or relay chain through the channel, and the second transaction is executed by that chain in parallel. To prevent bugs in parallel execution, Polkadot sets the source field in the transaction to identify the parallel chain; the cross-chain transaction fee can be discerned based on the logic of the transaction between parallel chains.

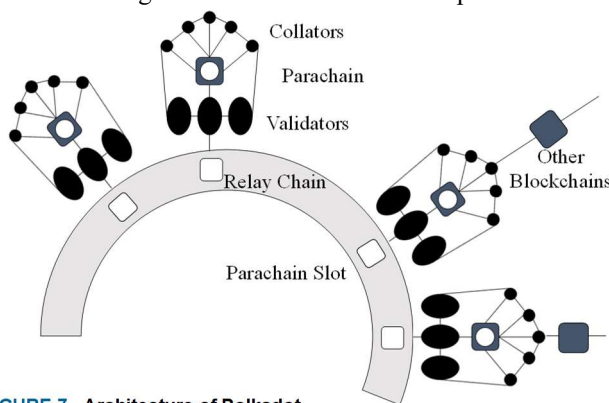


FIGURE 7. Architecture of Polkadot

In the Polkadot network, four participants work together to maintain the security and availability of the inter-chain network:

1) VALIDATOR

The validator has the highest authority, the validator needs to validate the nominated parallel chains involved in the new blocks generated, only when the validation is successful, the validator will link all the parallel chains involved in the block header to the relay chain blocks and carry out consensus. The validator needs to pledge sufficient tokens deposit to the system, which may include a portion of the nominator's deposit, in order to improve the security of the system in this way.

2) COLLATOR

Responsible for assisting the verifier to collect, verify, and submit alternative blocks and run a full node on a specific parallel chain, collect and execute transactions, and send new

blocks, send unencrypted blocks to the relevant verifier using zero-knowledge proofs [76].

3) FISHERMAN

Fisherman is not involved in any block packing and sending work, but only responsible for illegal transactions and invalid blocks to report, Fishermen find problems and report successfully will reap large rewards, Fisherman's report can achieve the supervision of the verifier. In the Polkadot network, Fishermen have limited access but high demand, so it is less difficult for Fishermen to enter, and there are no high restrictions on deposit, computing power and online time, but the reward for successful prosecution is huge, so the Polkadot network can efficiently maintain on-chain security with this incentive mechanism.

4) NOMINATOR

Nominator has the right to choose and trust the validator in the system and use the validator as their representative. Nominator will deliver his assets as a deposit to the validator, and the validator will deliver to the system, and the delivery will be rewarded and punished according to the success of the validator (and the selected validator will be rewarded and punished according to the rate of deposit), so it has certain risks and benefits.

The process of cross-chain transactions in Polkadot can be summarized as external transactions are sent to the inside of Polkadot network, Collator verifies the validity of the transaction and then packages the transaction information, and after Fisherman checks to exclude malicious attacks, the transaction is sent to the parallel chain, verified by the block header data in the relay chain, and then Validator verifies and packages the transaction to be transferred the transaction is sent to another parallel chain.

Polkadot uses the WebAssembly protocol to independently deploy network upgrades. To meet the increasing demand, without the risk of network forks, chain upgrades can be implemented without hard forks.

Polkadot supports interoperability based on state transition verification, which is done by a chain relay verifier. Parallel chains communicate through Cross-chain Message Passing Protocol (XCMP), a queueing communication mechanism based on Merkle trees [82]. Proof of state transition from parallel chains to relay chains is achieved by erasure coding (EC).

C. LIGHTNING NETWORK

Lightning Network [83,84] is a concept proposed by Joseph Poon and Thaddeus Dryja, which is based on micro-payment channel technology, and belongs to the category of state channel technology, whose core algorithm is to transfer the on-chain transactions to the off-chain, rely on the channel technology to achieve asset transfer, and move back to the chain for identification when problems occur. The security of the chain ensures that the two sides of the transaction under the chain will not cheat, thus realizing the expansion of the chain.

Lightning Network is designed for micro-payment transaction scenarios and is characterized by omitting the intermediate process of on-chain transactions, recording only the final state of both parties after the transaction on the chain, and recording a large amount of transaction information and processes off-chain. The Lightning Network's transaction method started as a solution to the low throughput of Bitcoin's off-chain scaling and has since evolved into an off-chain transaction method that enables cross-chain transactions between different chains.

This cross-chain transaction method is to establish the corresponding payment channel to complete the transaction, the transaction is completed to close the corresponding channel and according to the off-chain ledger for settlement, after the settlement is completed, the final state data of both parties on the chain. The above method is the initial version of Lightning Network, later to further enhance the transaction speed, allowing transactions through the original transaction channel, such as the existence of channels between Bitcoin chain A and Ethereum chain B, Ethereum chain B and Bitcoin chain C, A and C can be lightning payment transactions through the way A to B and then B to C. After a certain period of accumulation, the payment channel will form a large routing network, and as long as a user is connected to the network, he can realize the off-chain channel payment with all nodes of the network, which is called the payment channel network.

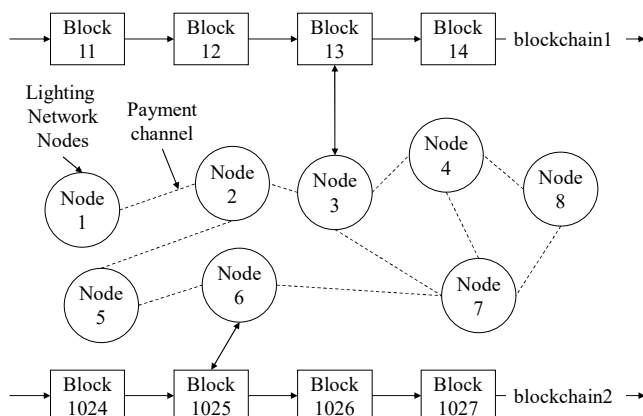


FIGURE 8. Off-chain payment network

The node that keeps the payment network online at all times and keeps its own channel open can be considered as an "off-chain miner", and the off-chain miner can charge a fee for using the channel, which is customized by the miner, and the user can use the Bellman-Ford-Gibson algorithm [85,86] to obtain the shortest distance (both the fee and the number of hops are sought to be small [87]). With the above incentives, the lightning payment technology has formed a very large off-chain payment network. Figure 8 shows a schematic diagram of the off-chain payment network, with transaction data uploaded to block 13 of chain 1 and block 1025 of chain 2.

Lightning Network ensures off-chain payment security by using RSNC and HTLC contracts.

1) HTLC

HTLC contract is hashed time lock contract, using this contract to achieve time-limited transfer, both sides of the transaction need to pre-deposit frozen funds, through each other's signature password to unlock the transaction funds, the implementation process and principle of hash lock technology.

2) RSMC

RSMC contract is the recoverable sequence maturity contract, the use of the pool mechanism to control the withdrawal process between the two sides, the parties need to pre-deposit a certain amount and freeze, and ensure that the amount of each transaction shall not exceed the maximum amount of pre-deposit, after each transaction, the fund status and distribution result shall be signed and confirmed, and update the original trading plan, which must be signed by both parties to be legal. If either party uses the invalidated or illegal scheme, the other party can challenge it, and if the challenge is successful, the pre-frozen amount will be delivered to the challenging party, the party that withdraws early due to a breach of contract will also be required to pay a default fee to the other party as compensation. The RSMC contract guarantees the security and durability of offline transactions and further guarantees the security and existence of the payment channel.

Lightning Network uses multiple signatures to lock the pre-stored token of transaction, and both parties jointly control the token through signatures to ensure the atomicity of the transaction. The Lightning Network sponsors are implementing AMP multi-path atomic payments, with the goal of making them suitable for large asset cross-chain transactions with short latency and low fees and improving the security of the payment process.

D. FUSION

The Fusion [88] project is developed by the Fusion Foundation, based on the DCRM (Distributed Control Rights Management), which is the process of mapping control of digital assets in the hands of an individual or organization to be managed by decentralized blockchain. Its main goal is to provide support for cross-chain transactions between heterogeneous blockchain systems, including the Fusion blockchain, and financial systems, with transactions including token, data information and off-chain information.

Fusion uses distributed nodes to control heterogeneous chain private keys and provides API interfaces for inter-chain calls, which are implemented using smart contracts, so that Fusion can be considered as a large and highly convergent cross-chain smart contract platform, the development team uses timer-triggered and event-triggered methods to control smart contracts in order to meet cross-chain application scenarios. Timer-triggered means that the smart contract is controlled by time conditions, for example, a loan transaction is controlled by a certain length of time, within which the lender needs to repay the loan, otherwise the smart contract executes an operation to ensure the interests of the borrower; or the smart contract executes an operation after the stock

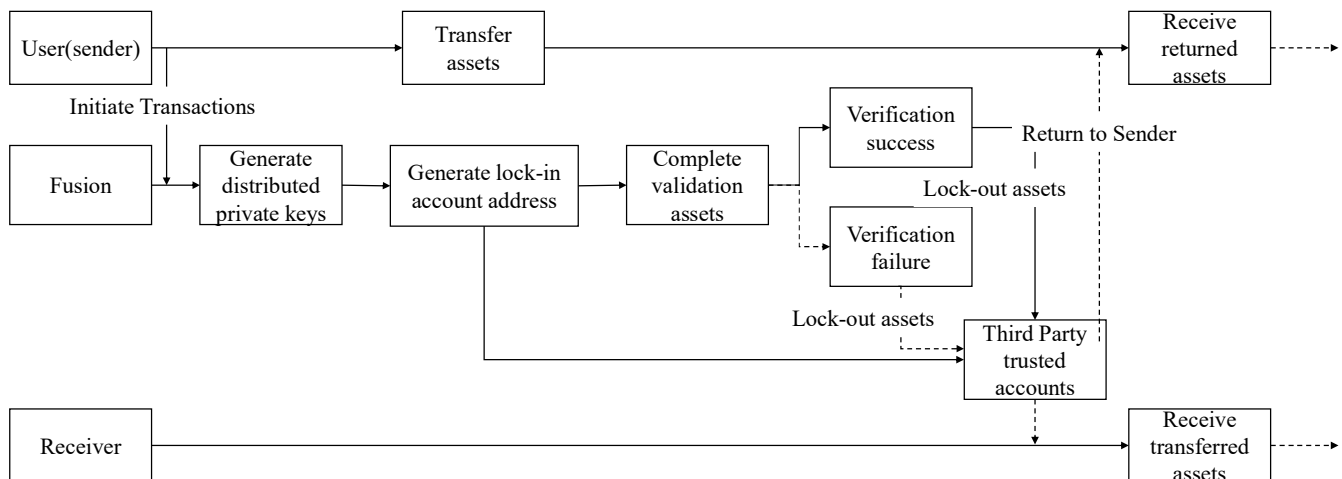


FIGURE 9. Fusion Asset Transfer Process

position operation reaches a certain point in time, according to the time point set by the user to cover or lock the position. Event-triggered method is the smart contract is controlled by events, such as loan transactions, the lender early repayment that triggers the smart contract to end the transaction.

Fusion controls asset management based on distributed private key technology, divides asset ownership and usage rights, and controls assets by a non-centralized system, specifically by Lock-in and Lock-out. Lock-in, that is, distributed generation of private keys, and then the system will transfer the assets to a designated third-party trusted account, and verify the validity of its assets, control the assets, and inform the user of the asset custody address. Lock-out, after initiating a transaction Fusion system verifies the validity of the asset, and upon successful verification, releases the control of the asset and sends the asset from the escrow account to the recipient, and if verification fails, returns the asset to the sending user, the locking and unlocking process of the asset is shown in Figure 9.

Cross-chain smart contracts are the core of the Fusion project. The various blockchains in the platform will have their private keys managed by the distributed nodes of the platform and will map their own tokens to the core Fusion chain to achieve cross-chain operations, we can understand Fusion chain as a large relay chain with its own token (FSN coin). However, unlike the relay method, a single node is not allowed to keep the complete escrow private key and must split the private key into N copies in accordance with the technical requirements of distributed signature notary, which is kept by N nodes to ensure decentralization and security within the platform.

E. INTERLEDGER

The Interledger [89] project is developed by the Ripple project team, based on a notary authentication algorithm, to provide cross-ledger payment functionality. Interledger creates a system that provides an interconnection between two parties to a transaction, which allows both parties to transact across

chains through third-party connectors without establishing trust. The connectors compete with each other to provide the best transaction efficiency and lowest cross-chain fees for users, and the chains in which the connectors are located allow for highly adaptable payment scenarios between any ledgers. Both parties to a transaction use asymmetric cryptography to escrow locked funds through their respective ledgers to allow secure cross-chain payments that rely on untrusted connectors.

The Interledger protocol provides two transaction methods [89]:

3) ATOMIC METHOD

Atomic method, in which both parties to a transaction choose a specific set of notaries to participate in the funds transfer payment process, provides transaction atomicity for payments.

4) GENERIC METHOD

Generic method, which uses a limited execution window, participant incentives, and "reverse" execution to ensure that both parties can complete a secure payment without trusting a third party, as an alternative to a notary public, uses incentives that do not share any trust.

Interledger uses a cryptographic escrow approach [90] to avoid any authentication failures or malicious node attacks in cross-chain payments by isolating all participants in the cross-chain and uses a two-stage submission protocol to enable secure intermediate payments. Cryptographic signatures secure the funds, Interledger uses asymmetric cryptography to hide the escrowed funds and needs to wait for a valid public key or hash signature to decrypt the escrowed funds.

The Interledger atomic method uses a multiple notary, as the multiple notary can customize the signature ratio to decide whether to approve cross-chain transactions, atomic method is lighter and faster, but can have security issues. Therefore, Interledger uses Byzantine fault-tolerance protocols to manage and incentivize notaries to properly verify that up to 33% of Byzantine malicious nodes are allowed.

Interledger also uses timer method to prevent funds from being permanently held in an escrow account, which returns

funds to the original account after the transaction process has exceeded the maximum allowable time, noting that the timeout method is used to return escrowed funds even in the event of a failed transaction.

Interledger ensures that both parties can conduct cross-chain transactions without trust, so the core of the protocol is to standardize the address of each ledger account so that the system can quickly address and verify the validity and security of the account. At the same time, it standardizes the information transmission format of cross-chain transactions, requiring both parties to the transaction and the third-party notary to use the same information format.

F. ETHER UNIVERSE

Ether Universe is developed by EOSIO and uses notary and sidechain technology. The notary scheme enables fast asset exchange, while the sidechain improves communication efficiency. DPOS consensus is adopted, and miners and guarantors are introduced to participate in verification and signature together with the notary scheme to improve system security. At the same time, 48 nodes are deployed, using distributed node storage and accounting technology to receive snapshots of user transactions across the chain in parallel and in real time to synchronize and verify the data for the transactions. The validity of the transaction is confirmed based on the verification of the transaction snapshot by each node of the Ether Universe node network. Although the verification time is about 150s, the distributed universe nodes ensure the security and validity of the transaction within the network as much as possible. The Ether Universe network can act as an intermediary for inter-chain transactions and can be considered as a routing or relay chain.

In terms of security, Ether Universe uses hot and cold wallet [91] technology to guarantee the safety of funds on the chain. Cold wallets are offline wallets, which cannot be accessed by malicious attacks or hackers. Ether Universe deploys 48 Universe nodes at the same time, and as long as 25(51%) of them are still working properly, the whole system can operate normally, so even if a single node or a few nodes are hacked, the system will not collapse. Moreover, the individual nodes of the Ether Universe are so secure that they can withstand malicious network attacks of up to 13GB/S.

G. ANTCHAIN

Antchain [92] cross-chain data connection service ODATS [93] (Open Data Access Trusted Service) supports the trusted interaction of data between homogeneous/heterogeneous blockchains, and its technical feature is to develop a full-stack cross-chain protocol called UDAG. The UDAG realize the trusted interaction between homogeneous or heterogeneous blockchains through the protocol stack and cross-chain contract in the underlying blockchain.

Antchain has four main cross-chain functions: cross-chain mutual recognition data specification, Ant UDAG cross-chain

protocol, cross-chain contract service, and TEE-based Oracle clustering service.

The UDAG cross-chain protocol ensures trustworthy and secure data transmission, UDAG includes smart contract cross-chain communication protocol and identity protocol: the cross-chain communication protocol provides a cross-chain communication interface between programmable smart contracts, providing a reliable orderly/disorderly communication protocol similar to TCP/UDP [94], the communication protocol is based on trust-neutral data anchoring and network routing, aiming to establish a layer of information communication protocol for smart contracts published on different chains, so that the cross-chain message communication of smart contracts on the blockchain is secure and reliable, and includes the trusted transmission of external data sources to the chain. Identity protocols allow businesses to securely cross-chain in an Internet of value consisting of blockchains by designing blockchain certificates that describe the unique authentication root of the blockchain as well as its secure update and public revocation. The cross-chain contract service provides smart contract-based cross-chain service capability to federated chain users by creating a domain name for the blockchain when registering the blockchain, which will be the unique identification of the blockchain when communicating across the chain. And cross-chain authorization communication requires authorization from the other blockchain before communication can start, and the authorization includes ledger data authorization and contract message authorization. The ledger data includes blockchain transactions, blocks and block headers, when authorizing other blockchains to get the ledger data of the current blockchain, user chain can specify which type of ledger data and contract messages to authorize. After cross-chain authorization of contract messages, the user blockchain can accept remote push contract messages from the authorized blockchain to perform complex interoperability and realize various business scenarios.

The Antchain cross-chain data connection service supports Ledger Data Access and Contract Message Pushing. Ledger Data Access is the target blockchain can directly access the relevant data of the source blockchain through the cross-chain service. Contract message pushing is the source blockchain can send messages to the smart contract of the target blockchain through the cross-chain service, and the smart contract of the target chain will finish processing the messages.

Through its cross-network deployment technology, Antchain deploy blockchain nodes across the cloud platform according to the requirements, that is, some nodes participating in the consensus run on the platform, and some nodes participating in the consensus run on the user's IT environment to achieve Scale out. Antchain uses the parallel consensus technology and multiple independent blockchain nodes are deployed in the network, ensuring network security and improving system response speed to achieve the Scale up of scalability.

The advantage of Antchain is to realize heterogeneous compatibility, support interconnection and interoperability of consortium chains, private chains and public chains, and verify blockchain data by TEE [95,96] and zero-knowledge proof to reduce the complexity of cross-chain authentication and improve the efficiency of cross-chain communication. In terms of cross-chain contract execution, Antchain supports users to authorize specified contracts of other blockchains, push cross-chain contract messages, and achieve remote invocation of contracts after cross-chain addressing, which greatly improves cross-chain interoperability and scalability.

H. BTC RELAY

BTCRelay [25,97] is developed by the ConsenSys team, aiming to enable the invocation of funds on the Bitcoin chain in the Ethereum environment. BTCRelay uses smart contract technology to validate transactions on the Bitcoin chain and provide relevant information to the Ethereum, enabling users to make payments with Bitcoin on the Ethereum chain.

BTCRelay implements the embedded SPV Two-way Peg technology in the process of using smart contracts on Ethereum, and includes a small bichain in the smart contract, and keeps simplified bichain transaction information (block header hash value), this approach is widely used in a large number of subsequent cross-chain projects, when the user needs to use the funds on the bichain, the smart contract can use the block header data and merkle-tree path to verify the legitimacy of the transaction.

The disadvantages of this method include excessive reliance on other members of the BTCRelay environment, high fees, and the growing volume of contracts due to the

continuous nesting of internal smart contracts, so this method is now in a semi-deprecated state.

BTCRelay kernel technology is smart contract and cannot be classified as cross-chain technology according to current standards, but it is considered the first cross-chain invocation in the history of blockchain and the first sidechain of blockchain, which has epochal significance and laid a good foundation for the subsequent attempts and development of cross-chain technology.

I. COMPARISON OF BLOCKCHAIN PROJECTS

Table 2 compares cross-chain projects based on their consensus mechanisms, implementation difficulties[98], cross-chain technologies used, security[25], flexibility[98], and speed of cross-chain transactions, for example, the cross-chain process of Cosmos, Polkadot and Interledger can be summarized as four steps: establishing connection, establishing transmission channel, validation and feedback. However, the way to implement the transmission channel, verification and feedback varies, so their performance focuses on security, flexibility and transaction speed. The table shows that cross-chain mainstream projects are mainly supported by the above four cross-chain technologies, such as Cosmos' IBC protocol based on relay chain technology for improvement. It is worth noting that Polkadot uses four consensus schemes, asynchronous BFT, NPoS, BABE and GRANDPA, among which asynchronous BFT asynchronous Byzantine is used in the relay chain as the main technology. From the table, it is easy to find that the three attributes of security, flexibility, and transaction speed conflict, which is determined by the consensus scheme and security mechanism used by some projects.

TABLE 2. Comparison of cross-chain projects

Cross-chain project	Consensus	Implementation difficulty	Cross-chain technology	Security	Flexibility	Transaction speed	Transaction type
Cosmos	Tendermint BFT	Low	IBC Protocol	Medium	High	Very High	Smart Contract
Polkadot	Asynchronous BFT	High	Relays	High	Low	Medium	Smart Contract
Lighting Network	Following transactions chain	Low	Channels	Medium	Medium	High	Off-line transfer
Fusion	PoW	Medium	DCRM	Medium	Medium	Medium	transfer
Interledger	Following transactions chain	Low	Multi-Notary	Low	High	Medium	Smart Contract
Ether Universe	DPoS	Medium	Notary and Sidechains	High	Medium	Low	Smart Contract
Antchain	Following transactions chain	Medium	UDAG Protocol	High	High	High	Smart Contract
BTC Relay	PoW	Low	SPV	High	Low	Very Low	Smart Contract

V. FUTURE RESEARCH TREND OF CROSS-CHAIN

Blockchain cross-chain technology is still in its initial stage, but in the future, when blockchain technology penetrates into various industries, cross-chain technology will provide value and information interaction between industry chains. The

further development of cross-chain technology will continue to promote the integration of industries, promote the prosperity of digital assets and help traditional industries to innovate, so that the real economy can flourish under the stimulation of the Internet. However, cross-chain technology still needs to consider challenges including system reliability,

interactivity, transaction concurrency control, query optimization and security.

A. CROSS-CHAIN PROCESSING RELIABILITY

Cross-chain processing reliability refers to the impact of cross-chain transaction on both sides of the transaction, the success rate of the transaction and the transaction processing ability after the failure of the cross-chain transaction. However, many cross-chain projects are handled in a negative way under the condition of cross-chain failure. For example, in the case of transaction failure, Fusion project will return the escrow funds in the way of timeout. If the node of another transaction party fails or goes offline, it needs to wait for a long time before the funds can be returned. The above method is not conducive to the reliability of inter-chain transactions, so it is necessary to find a reliable method to solve the inter-chain communication to avoid node failure and other emergencies.

In addition, the cross-chain operation is limited by the third-party nodes of the third-party notary scheme or the off-chain state transaction channel of the lightning network, which will greatly affect the success rate of cross-chain transactions once the third-party notary network is congested or the off-chain state channel fails. For example, the SPV Two-way Peg technology used by Cosmos requires locking the assets of both parties, and if the assets of one-party change, the transaction will be abandoned to prevent malicious double-spend attacks, which may lead to long waiting time or transaction failure for both parties due to network congestion. Even though the atomicity mechanism of cross-chain asset transactions can guarantee the security of assets, the atomicity mechanism involving cross-chain data flow cannot guarantee its effectiveness, so a highly reliable and efficient cross-chain integrated solution is needed to deal with unexpected situations and ensure the reliability of cross-chain processing.

The current consortium chain projects are mostly developed based on RAFT and PoA (Proof of Authority) consensus protocols, whose Byzantine node tolerance level is only 33%, and even Ripple consensus protocol tolerance level is only 20%, and the large-scale applications are likely to result in busy cross-chain verification nodes leading to untimely signatures, while repeated cross-chain access applications by cross-chain interaction nodes are likely to cause large-scale network congestion or even paralysis. Therefore, how to improve the fault tolerance efficiency on the federated chain [99] to ensure the fast and effective cross-chain transmission of non-confidential information for future large-scale applications of federated chain has become one of the possible research directions in the future.

B. CROSS-CHAIN INTERACTIVITY

In order to realize the deep industrial integration based on blockchain, the first thing that blockchain technology achieves is data interconnection and interoperability, but the security mechanism and data privacy mechanism of blockchain make cross-chain information exchange a major problem. Most of

the current cross-chain projects are still aimed at inter-chain asset exchange and lack practical data communication research. To ensure reliable and efficient information sharing, it will be a future research problem to achieve efficient communication transmission without destroying the original data privacy of the blockchain and the security of the blockchain ecosystem.

Cross-chain interactivity also involves the difference between public chains and consortium chains. The transmission of information between public chains needs to consider the impact of the bottleneck effect on the communication efficiency caused by the consensus on the chain, and whether the possible expansion plan of public chains in the future can realize the efficiency and trustworthiness of information transmission of public chains. The consortium chain needs to consider the permission control scheme of information transmission under the regulatory conditions, how to encrypt the transmission of unauthorized nodes between chains, and whether the intermediate trusted nodes of cross-chain transmission on the consortium chain need to review the transmission information. In the future, we also need to consider the research of cross-chain transmission of information between the consortium chain and the public chain, such as the cross-chain interaction between a tax consortium chain supervised by the government and the public chain of the online shopping platform, and the nodes of the consortium chain can retrieve the information of the public chain according to the authority to facilitate tax management. Whether the public chain can exist in the form of "node on the consortium chain" or sidechain to realize cross-chain interaction, as well as the authority management and information transmission of the consortium chain to the public chain, are hot issues for future research.

With the development of blockchain technology and social needs, a large number of federation chain applications will be implemented soon, and non-confidential cross-chain information transmission will become one of the main applications in a large-scale application environment. It should be noted that cross-chain transmission needs to be carried out in an orderly manner under the supervision of the consortium chain to prevent data tampering. The original on-chain consensus protocol should not be broken, and the access node permission scheme should be observed, and only cross-chain fault tolerance should be improved to prevent the leakage of "precious" information on the consortium chain.

C. CROSS-CHAIN TRANSACTION CONCURRENCY CONTROL

In order to improve the performance of blockchain and reduce the burden of main chain to achieve functional diversification, cross-chain transaction concurrency technology will become one of the future research hotspots, and good subtransaction division and concurrent execution of sidechain transactions will greatly improve the scalability of on-chain systems.

Subtransaction partitioning can be understood as the chain system distributes multiple transactions committed to multiple chains for concurrent execution, and concurrent execution requires the study of transaction isolation and parallel result conflicting issues to ensure the ultimate consistency of transactions.

At present, the concurrency of cross-chain transactions, such as the Sidechains/Relays scheme, relies on inter-chain smart contracts, which may lead to congestion in transaction processing as the volume of transactions increases, resulting in downtime of the on-chain system, and future research can focus on efficient and reliable control strategies for inter-chain transactions.

Inter-chain transaction concurrency is also limited by the possibility of different consensus protocols and cross-chain protocols between systems, which makes transaction concurrency development difficult and implementation complex. Cross-chain research can be based on developing standardized interfaces and matching various consensus protocols to facilitate cross-chain transaction concurrency and improve inter-blockchain system adaptability.

D. CROSS-CHAIN QUERY OPTIMIZATION

Blockchain query [100] research can be divided into general query processing and trusted query processing. general query processing includes studying the efficiency of data traceability query in blockchain environment, which can be achieved through graph topology [101], block data index and view [102], on-chain query statements, etc., and also includes smart contracts that support data traceability. Trusted query processing studies the trustworthiness of query results based on the condition that the blockchain environment is not trustworthy, and ensures the trustworthiness and efficiency of query results through hash values, prefix trees, block indexes, etc.

In the cross-chain scenario, query processing is mostly based on untrustworthy environment, and it is difficult to trace the source of query processing, which involves various issues such as trustworthiness, efficiency, and overhead, etc. The large number of smart contracts generated by large-scale query processing may cause network congestion or even system collapse. Therefore, cross-chain query optimization can be one of the future research directions, with the goal of efficient and trustworthy query processing and reduced query and verification overhead.

Cross-chain query optimization can improve the efficiency of query and validation by building intra-chain as well as inter-chain indexes, and blockchain cross-chain query statements [103,104] can be studied to realize the operations of table building, insertion, and selection of blockchain relational tables in cross-chain to improve the efficiency of cross-chain query.

E. CROSS-CHAIN SECURITY

The existing blockchain cross-chain technology has still not been applied on a large scale, and the important reason is that the current cross-chain security issues are still not guaranteed, and the security issues include [25] trust dependency, malicious transactions, and chain structure influence.

For example, the Sidechains/Relays scheme relies on the block header data for verification, which cannot fully verify the transactions by obtaining all transaction data of the network as the single chain or the main chain and is vulnerable to malicious attacks such as double-spending. The notary scheme relies on third-party organizations or institutions, and even with the use of distributed notary scheme and good election strategy, the cross-chain interaction of information and assets still requires the honesty of third-party notaries, and there is a risk of complicity [25], which gives up the decentralized nature of blockchain technology and causes the centralized dependence of cross-chain transactions. Hash-locking scheme relies on time lock, such as a malicious attack in lightning network system can cause massive system congestion by creating a large number of transaction channels and transactions in the system and making the channel timeout, and the lightning network system congestion may generate the risk of user funds being stolen.

In the event of a hard fork of a blockchain system, the same key, transaction address, and chain structure may lead to a cross-chain replay attack, where a transaction from one chain is broadcast to multiple chains and the transaction is confirmed multiple times. Cross-chain Replay Attacks may lead to a large loss of user assets and information. A long delay or congestion in cross-chain transactions may also lead to Race Conditions Attack [105], which results in loss of assets for a cross-chain participant.

Cross-chain systems may involve multiple blockchains with different designs of structure, consensus and compatibility, which may lead to various security issues in cross-chain asset transactions and cross-chain information interactions, and future research needs to take cross-chain security as the basis and premise.

F. POTENTIAL APPLICATIONS

In the P2P model, as the number of bridged blockchains increases, the number of smart contracts required to maintain the operation of the cross-chain bridge also increases quadratically. Smart contracts written at different times are also rapidly increasing security risks. Therefore, more secure cross-chain bridge applications will be needed in the future. Wormhole and Nomad projects, for example, promised to add contract configuration checks in the next release after the attacks. Wormhole is also developing three security features: regulatory, accounting, and emergency shutdown, which will greatly improve the project security.

On-chain data security has been successful, but data Cross-Chain interaction is still a promising future application. Currently, Oracle Blockchain Platform and HUAWEI TCS (Trusted cross-chain Service) have proposed data cross-chain

services. Although the existing cross-chain data services have limited functions and are often applied in the financial field, they will be applied in the medical, judicial and education fields in the future.

VI. CONCLUSION

The current blockchain technology is developing rapidly. Cross-chain technology, as a key interoperability technology [98] for the integration and development of future blockchain systems. In this paper, we surveyed the main technologies and projects involved in cross-chain technology in blockchain technology, compared the existing improved algorithms with the mainstream project solutions and provided an outlook on the current development trend and the problems to be solved. Although the existing cross-chain technology is not perfect, the demand for cross-chain from different perspectives, application scenarios and technologies will increase significantly in the future and the key technologies for cross-chain are still in the blank period. Blockchains with different architecture will be generated continuously, so a blockchain world with multiple chains coexisting will be created soon. With the development of cross-chain technology, it will be applicable to more blockchains with different structures and application scenarios. Cross-chain technology will further improve the access speed and query efficiency of blockchains, reduce the occurrence of errors and improve the security of blockchain system. Despite many problems and challenges to be overcome, the development of cross-chain technology will certainly have a revolutionary impact on the blockchain world.

REFERENCES

- [1] Y. Ge, T. Nie, X. Li, D. Shen, and Y. Bao, "The Challenge and Prospect of Distributed Data Management Techniques in Blockchain Systems," *Chinese Journal of Computers.*, vol. 1, no. 2, pp. 28-54, 2021.
- [2] P. He, G. Yu, Y. Zhang, and Y. Bao, "Survey on Blockchain Technology and Its Application Prospect," *Computer Science.*, vol. 04, no. 17, pp. 7-15, 2017.
- [3] H. Seike, Y. Aoki, and N. Koshizuka, "Fork Rate-Based Analysis of the Longest Chain Growth Time Interval of a PoW Blockchain," *2019 IEEE International Conference on Blockchain. (Blockchain)*, 2019, pp. 253-260.
- [4] T. Keenan, "Alice in Blockchains: Surprising Security Pitfalls in PoW and PoS Blockchain Systems," *2017 15th Annual Conference on Privacy, Security and Trust. (PST)*, 2017, pp. 400-402.
- [5] C. Ma, "Research on Improved Consensus Mechanism of Delegated Proof of Stake in Blockchain," *Liaoning University*, 2020.
- [6] Y. Luo, Y. Chen, Q. Chen, and Q. Liang, "A New Election Algorithm for DPos Consensus Mechanism in Blockchain," *2018 7th International Conference on Digital Home. (ICDH)*, 2018, pp. 116-120.
- [7] R. Nakamura, T. Jimba, and D. Harz, "Refinement and Verification of CBC Casper," *2019 Crypto Valley Conference on Blockchain Technology. (CVCBT)*, 2019, pp. 26-38.
- [8] E. Li, T. Serbenut, D. Diaconescu, V. Zamfir, and G. Rosu, "Formalizing Correct-by-Construction Casper in Coq," *2020 IEEE International Conference on Blockchain and Cryptocurrency. (ICBC)*, 2020, pp. 1-3.
- [9] V. Buterin, D. Reijersbergen, S. Leonardos, and G. Piliouras, "Incentives in Ethereum's Hybrid Casper Protocol," *2019 IEEE International Conference on Blockchain and Cryptocurrency. (ICBC)*, 2019, pp. 236-244.
- [10] H. Sukhwani, J. Martinez, X. Chang, K. Trivedi, and A. Rindos, "Performance Modeling of PBFT Consensus Process for Permissioned Blockchain Network (Hyperledger Fabric)," *2017 IEEE 36th Symposium on Reliable Distributed Systems. (SRDS)*, 2017, pp. 253-255.
- [11] G. Yu, B. Wu, and X. Niu, "Improved Blockchain Consensus Mechanism Based on PBFT Algorithm," *2020 2nd International Conference on Advances in Computer Technology, Information Science and Communications. (CTISC)*, 2020, pp. 14-21.
- [12] K. Wang, "Research and Application of Consortium Blockchain Security and Performance," *Beijing University of Technology.*, 2020.
- [13] A. Charapko, A. Ailijiang, and M. Demirbas, "Bridging Paxos and Blockchain Consensus," *2018 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data. (SmartData)*, 2018, pp. 1545-1552.
- [14] D. Huang, L. Li, B. Chen, and B. Wang, "RBFT: a new Byzantine fault-tolerant consensus mechanism based on Raft cluster," *Journal of Communications.*, vol. 3, no. 42, pp. 209-219, 2021.
- [15] S. Alrubei, E. Ball, and J. Rigelsford, "Securing IoT-Blockchain Applications Through Honesty-Based Distributed Proof of Authority Consensus Algorithm," *CyberSA.*, 2021, pp. 1-7.
- [16] V. Clincy, and H. Shahriar, "Blockchain Development Platform Comparison," *2019 IEEE 43rd Annual Computer Software and Applications Conference. (COMPSAC)*, 2019, pp. 922-923.
- [17] F. Ghaffari, E. Bertin, J. Hatim, and N. Crespi, "Authentication and Access Control based on Distributed Ledger Technology: A survey," *2020 2nd Conference on Blockchain Research & Applications for Innovative Networks and Services. (BRAINS)*, 2020, pp. 79-86.
- [18] Hyperledger, "Hyperledger Fabric White Paper," 2020. <https://hyperledger-fabric.readthedocs.io/en/release-2.4/whatis.html>
- [19] P. Ruan, D. Loghin, Q. Ta, M. Zhang, G. Chen, and B. Ooi, "A Transactional Perspective on Execute-order-validate Blockchains," *SIGMOD.*, 2020, pp. 543-557.
- [20] C. Zhuang, Q. Dai, and Y. Zhang, "BCPPT: A blockchain-based privacy-preserving and traceability identity management scheme for intellectual property," *Peer-to-Peer Netw.*, 2022, pp. 724-738.
- [21] S. He, X. Huang, and X. Chen, "The Research Summary of the Development and Application of Blockchain Cross-chain Technology," *Journal of Xihua University. (Natural Science Edition)*, vol. 03, no. 1, pp. 1-14, 2021.
- [22] A. Lu, K. Zhao, J. Yang, and F. Wang, "Research on Cross-chain Technology of Blockchain," *Netinfo Security.*, vol. 08, no. 07, pp. 83-90, 2019.

- [23] Z. Xu, and X. Zhou, "Survey on crosschain technology," *Application Research of Computers.*, vol. 02, no. 38, pp. 341-346, 2021.
- [24] Z. Guo, S. Guo, S. Zhang, L. Song, and H. Wang, "Analysis of cross-chain technology of blockchain," *Chinese Journal on Internet of Things.*, vol. 02, no. 4, pp. 35-48, 2020.
- [25] F. Li, Z. Li, and H. Zhao, "Research on the Progress in Cross-chain Technology of Blockchains," *Journal of Software.*, vol. 06, no. 30, pp. 1649-1660, 2019.
- [26] S. Lin, Y. Kong, and S. Nie, "Overview of Block Chain Cross Chain Technology," *2021 13th International Conference on Measuring Technology and Mechatronics Automation. (ICMTMA)*, pp. 357-360, 2021.
- [27] C. Fan, S. Ghaemi, H. Khazaei, and P. Musilek, "Performance Evaluation of Blockchain Systems: A Systematic Survey," *IEEE Access.*, vol. 8, pp. 126927-126950, 2020.
- [28] X. Zhang, H. Wang, P. Shi, and X. Fu, "LS4BUCC: A Low Overhead Storage Architecture for Blockchain Based Unmanned Collaborative Cognition System," *SOSE.*, 2019.
- [29] Q. Wang, H. Wang, and B. Zheng, "An efficient distributed storage strategy for blockchain," *ACM TUR-C.*, pp. 54:1-54:5, 2019.
- [30] J. Xie, F. Yu, T. Huang, R. Xie, J. Liu, and Y. Liu, "A Survey on the Scalability of Blockchain Systems," *IEEE Netw.*, vol. 05, no. 33, pp. 166-173, 2019.
- [31] Z. Sun, X. Zhang, F. Xiang, and L. Chen, "Survey on Storage Scalability on Blockchain," *Journal of Software.*, vol. 06, no. 1, pp. 1-20, 2021.
- [32] R. Qiao, Y. Cao, and Q. Wang, "Traceability mechanism of dynamic data in Internet of things based on consortium blockchain," *Journal of Software.*, vol. 6, no. 30, pp. 1614-1631, 2019.
- [33] H. Mao, T. Nie, D. Shen, G. Yu, S. Xu, and G. He, "Survey on Key Techniques and Development of Blockchain as a Service Platform," *Computer Science.*, vol. 11, no. 2, pp. 4-11, 2021.
- [34] Hyperledger. 2022. *Hyperledger Fabric*.
<https://github.com/hyperledger/fabric/blob/release-2.4/docs/source/whatis.md>
- [35] M. Schinle, C. Erler, P. N. Andris, and W. Stork, "Integration, Execution and Monitoring of Business Processes with Chaincode," *2020 2nd Conference on Blockchain Research & Applications for Innovative Networks and Services. (BRAINS)*, 2020, pp. 63-70.
- [36] J. Poon, and T. Dryja, "The Bitcoin Lightning Network: Scalable Off-Chain Instant Payments," *Lightning Network.*, 2020.
<https://lightning.network/lightning-networkpaper.pdf>
- [37] V. Siris, D. Dimopoulos, N. Fotiou, S. Voulgaris, and G. Polyzos, "IoT Resource Access utilizing Blockchains and Trusted Execution Environments," *Global IoT Summit. (GloTS)*, 2019, pp. 1-6.
- [38] Bitcoin Wiki, "Hashlock," 2016, Technical Report.
<https://en.bitcoin.it/wiki/Hashlock>
- [39] Bitcoin Wiki, "Timelock," 2016, Technical Report.
<https://en.bitcoin.it/wiki/Timelock>
- [40] Bitcoin Wiki, "Hash Time Locked Contracts," 2016, Technical Report.
https://en.bitcoin.it/wiki/Hash_Time_Locked_Contracts
- [41] M. Belotti, Stefano Moretti, Maria Potop-Butucaru, and Stefano Secci, "Game Theoretical Analysis of Atomic Cross-Chain Swaps," 2020, Technical Report. <https://hal.archives-ouvertes.fr/hal-02414356>
- [42] P. Zappalà, M. Belotti, M. Potop-Butucaru, and S. Secci, "Game Theoretical Framework for Analyzing Blockchains Robustness," 2020, Technical Report. <https://eprint.iacr.org/2020/626.pdf>
- [43] V. Zakhary, D. Agrawal, and A. Abbadi, "Atomic Commitment Across Blockchains," *Proc. VLDB Endow*, 2020, pp. 1319-1331.
- [44] Y. Sun, L. Yi, L. Duan, and W. Wang, "A Decentralized Cross-Chain Service Protocol based on Notary Schemes and Hash-Locking," *SCC*, 2022, pp. 152-157.
- [45] R. Khalil, P. Moreno-Sanchez, A. Zamyatin, A. Gervais, and G. Felley, "Commit-Chains: Secure, Scalable Off-Chain Payments," 2018, Technical Report. <https://eprint.iacr.org/2018/642.pdf>
- [46] B. Dai, S. Jiang, D. Li, and C. Li, "Evaluation model of cross-chain notary mechanism based on improved PageRank algorithm," *Computer Engineering.*, vol. 02, no. 3, pp. 26-31, 2021.
- [47] S. Ebrahimi, P. Hasanizadeh, S. Aghamirmohammadali, and A. Akbari, "Enhancing Cold Wallet Security with Native Multi-Signature schemes in Centralized Exchanges," *CoRR.*, 2021.
- [48] C. Baum, B. David, and T. Frederiksen, "P2DEX: Privacy-Preserving Decentralized Cryptocurrency Exchange," *ACNS.*, 2021, pp. 163-194.
- [49] Z. Yin, B. Zhang, J. Xu, K. Lu, and K. Ren, "Bool Network: An Open, Distributed, Secure Cross-Chain Notary Platform," *IEEE Trans. Inf. Forensics Secur.*, 2022, pp. 3465-3478.
- [50] E. Scheid, P. Kiechl, M. Franco, B. Rodrigues, C. Killer, and B. Stiller, "Security and Standardization of a Notary-based Blockchain Interoperability API," *BCCA*, 2021, PP. 42-48.
- [51] M. Dehez-Clementi, J. Lacan, J. Deneuville, H. Asghar, and D. Kaafar, "A Blockchain-enabled Anonymous-yet-Traceable Distributed Key Generation," *2021 IEEE International Conference on Blockchain (Blockchain)*, 2021, pp. 257-265.
- [52] D. Kalyani, and R. Sridevi, "Robust distributed key issuing protocol for identity based cryptography," *International Conference on Advances in Computing, Communications and Informatics. (ICACCI)*, 2016, pp. 821-825.
- [53] Z. Zhao, G. Wu, W. Susilo, F. Guo, B. Wang, and Y. Hu, "Accountable identity-based encryption with distributed private key generators," *Inf. Sci.*, 2019, pp. 352-366.
- [54] B. Xu, X. Zhang, H. Cao, Y. Li and L. Wang, "HERMS: A Hierarchical Electronic Records Management System Based on Blockchain with Distributed Key Generation," *SCC*, 2022, pp. 295-304.
- [55] A. Back, M. Corallo, L. Dashjr, M. Friedenbach, Gregory Maxwell, Andrew Miller, Andrew Poelstra, Jorge Timón, and Pieter Wuille, "Enabling Blockchain Innovations with Pegged Sidechains," 2020, Technical Report. Blockstream.
- [56] P. Gai, A. Kiayias, and D. Zindros, "Proof-of-stake sidechains," *IEEE Symposium on Security and Privacy.*, 2019, pp. 139-156.
- [57] E. Kokoris-Kogias, P. Jovanovic, L. Gasser, N. Gailly, E. Syta, and Bryan Ford, "OmniLedger: A secure, scale-out, decentralized ledger via sharding," *IEEE Symposium on Security and Privacy.*, 2018, pp. 583-598.

- [58] A. Zamyatin, N. Stifter, A. Judmayer, P. Schindler, E. Weippl, and W. Knottenbelt, "A wild velvet fork appears! Inclusive blockchain protocol changes in practice," *Financial Cryptography Workshops.*, 2018, pp. 31–42.
- [59] A. Garoffolo, D. Kaidalov, and R. Oliynykov, "Zendoo: a zk-SNARK Verifiable Cross-Chain Transfer Protocol Enabling Decoupled and Decentralized Sidechains," Technical Report.
- [60] S. Ye, X. Wang, C. Xu, and J. Sun, "BitXHub: Side-relayChain Based Heterogeneous Blockchain Interoperable Platform," *Computer Science.*, vol. 06, no. 47, 2020, pp. 294–302.
- [61] A. Singh, K. Click, R. Parizi, Q. Zhang, A. Dehghantanha, and K. Choo, "Sidechain technologies in blockchain networks: An examination and state-of-the-art review," *Journal of Network and Computer Applications.*, vol. 149, no. 5, 2020.
- [62] V. Buterin, "R3 Report - Chain Interoperability," *R3 Corda.*, 2020, Technical Report. https://www.r3.com/wp-content/uploads/2017/06/chain_interoperability_r3.pdf
- [63] A. Kiayias, and D. Zindros. "Proof-of-Work Sidechains", 2018 Technical Report. <https://eprint.iacr.org/2018/1048.pdf>
- [64] P. Fraunthaler, M. Sigwart, C. Spanring, and S. Schulte, "Testimonium: A cost-efficient blockchain relay," *arXiv preprint.*, 2020, Technical Report. <https://arxiv.org/pdf/2002.12837.pdf>
- [65] P. Ray, N. Kumar, and D. Dash, "BLWN: Blockchain-Based Lightweight Simplified Payment Verification in IoT-Assisted e-Healthcare," *IEEE Systems Journal.*, vol. 15, no. 1, 2021, pp. 134–145.
- [66] S. Nakamoto. Bitcoin, "A peer-to-peer electronic cash system," 2008, Technical Report. <https://bitcoin.org/bitcoin.pdf>
- [67] Bitcoin Wiki, "Simplified Payment Verification (SPV)," 2020, Technical Report. https://en.bitcoinwiki.org/wiki/Simplified_Payment_Verification
- [68] M. Szydło, "Merkle tree traversal in log space and time," *International Conference on the Theory and Applications of Cryptographic Techniques.*, 2016, pp. 541–554.
- [69] D. Ding, K. Li, L. Jia, Z. Li, J. Li, and Y. Sun, "Privacy protection for blockchains with account and multi-asset model," *China Communications.*, vol. 16, no. 6, 2019, pp. 69–79.
- [70] W. Suliyanti, and R. Sari, "Evaluation of Hash Rate-based Double-Spending based on Proof-of-Work Blockchain," *International Conference on Information and Communication Technology Convergence. (ICTC)*, 2019, pp. 169–174.
- [71] G. Ramezan, C. Leung, and Z. Jane Wang, "A Strong Adaptive, Strategic Double-Spending Attack on Blockchains," *iThings/GreenCom/CPSCoM/SmartData.*, 2018, pp. 1219–1227.
- [72] Y. Li, J. Weng, M. Li, W. Wu, J. Weng, J. Liu, and S. Hu, "ZeroCross: A sidechain-based privacy-preserving Cross-chain solution for Monero," *J. Parallel Distributed Comput.*, 2022, pp. 301–316.
- [73] J. Kwon, and E. Buchman, "Cosmos Whitepaper," *Cosmos Foundation.*, 2021, Technical Report. <https://v1.cosmos.network/intro/what-is-tendermint-core-and-the-abci>
- [74] Tendermint, "Tendermint BFT," 2020, Technical Report. <https://github.com/tendermint/tendermint>
- [75] B. Assiri, and W. Khan, "Enhanced and Lock-Free Tendermint Blockchain Protocol," *IEEE International Conference on Smart Internet of Things. (SmartIoT)*, 2019, pp. 220–226.
- [76] IBC Ecosystem Working Group, "Inter-Blockchain Communication Protocol (IBC)," 2020, Technical Report. <https://github.com/cosmos/ics/tree/master/ibc>
- [77] J. Burdges, A. Cevallos, P. Czaban, R. Habermeier, S. Hosseini, F. Lama, H. Alper, X. Luo, F. Shirazi, A. Stewart, and G. Wood, "Overview of polkadot and its design considerations," 2020, Technical Report. <https://arxiv.org/abs/2005.13456>
- [78] G. Yu, B. Wu, and X. Niu, "Improved Blockchain Consensus Mechanism Based on PBFT Algorithm," *International Conference on Advances in Computer Technology, Information Science and Communications. (CTISC)*, 2020, pp. 14–21.
- [79] G. Wood, "Polkadot: Vision for a Heterogeneous Multi-Chain Framework," 2016, Technical Report. <https://github.com/w3f/polkadot-white-paper/raw/master/PolkaDotPaper.pdf>
- [80] A. Cevallos, and A. Stewart, "A verifiably secure and proportional committee election rule," *AFT.*, 2021, pp. 29–42.
- [81] M. Harikrishnan and K. Lakshmy, "Secure Digital Service Payments using Zero Knowledge Proof in Distributed Network," *International Conference on Advanced Computing & Communication Systems. (ICACCS)*, 2019, pp. 307–312.
- [82] Polkadot, "Cross-chain Message Passing (XCMP)," Polkadot Wiki, 2019, Technical Report. <https://wiki.polkadot.network/docs/en/learn-crosschain>
- [83] J. Wu, and S. Jiang, "Local Pooling of Connected Supernodes in Lightning Networks for Blockchains," *IEEE International Conference on Blockchain. (Blockchain)*, 2020, pp. 421–427.
- [84] Y. Guo, J. Tong, and C. Feng, "A Measurement Study of Bitcoin Lightning Network," *IEEE International Conference on Blockchain. (Blockchain)*, 2019, pp. 202–211.
- [85] M. Safari, W. Oortwijn, and M. Huisman, "Automated Verification of the Parallel Bellman-Ford Algorithm," *SAS.*, 2021, pp. 346–358.
- [86] Y. Mo, and L. Yu, "A Lyapunov analysis of the continuous-time adaptive Bellman-Ford algorithm," *Syst. Control. Lett.*, 2021, pp. 157: 105045.
- [87] C. Wang, "Research and implementation of blockchain lightning network," *University of Electronic Science and Technology of China.*, 2018.
- [88] Fusion Foundation, "Fusion Whitepaper: An inclusive Cryptofinance platform based on blockchain," 2017, Technical Report. <https://www.fusion.org/themes/fusion/assets/pdf/Fusion-White-Paper.pdf>
- [89] S. Thomas, and E. Schwartz, "A protocol for Interledger payments," 2015, Technical Report. <https://interledger.org/interledger.pdf>
- [90] Interledger, "Interledger Protocol V4 (ILPv4)," 2020, Technical Report. <https://interledger.org/rfcs/0027-interledgerprotocol-4/>
- [91] H. Xiao, and L. Tan, "Trusted protection scheme for virtual currency hot wallet," *Computer Engineering and Design.*, vol. 07, 2019, pp. 1801–1808.
- [92] Ant Group, "Antchain & Antchain Blockchain as a Service," 2019, Technical Report. <https://antchain.antgroup.com/docs/11/171879>
- [93] Antchain, "Antchain Open Data Access Trusted Service," 2022, Technical Report. <https://antchain.antgroup.com/docs/11/173098>

- [94] Y. Tsubouchi, M. Furukawa, and R. Matsumoto, "Low Overhead TCP/UDP Socket-based Tracing for Discovering Network Services Dependencies," *J. Inf. Process.*, 2022, pp. 260-268.
- [95] Y. Xia, Z. Hua, Y. Yu, J. Gu, H. Chen, B. Zang, and H. Guan, "Colony: A Privileged Trusted Execution Environment With Extensibility," *IEEE Trans. Computers.*, vol. 02, 2022, pp. 479-492.
- [96] L. Brenna, I. Singh, H. Johansen, and D. Johansen, "TFHE-rs: A library for safe and secure remote computing using fully homomorphic encryption and trusted execution environments," *Array* 13., 2022.
- [97] Ethereum Foundation, and Consensys, "BTC-relay: Ethereum contract for Bitcoin SPV," 2020, Technical Report. <https://github.com/ethereum/btcrelay>
- [98] R. Belchior, A. Vasconcelos, S. Guerreiro, and M. Correia, "A Survey on Blockchain Interoperability: Past, Present, and Future Trends," *ACM Comput. Surv.*, vol. 8, no. 54, 2022, pp. 168:1-168:41.
- [99] K. Lei, Q. Zhang, L. Xu, and Z. Qi, "Reputation-Based Byzantine Fault-Tolerance for Consortium Blockchain," *IEEE 24th International Conference on Parallel and Distributed Systems. (ICPADS)*, 2018, pp. 604-611.
- [100] Z. Zhang, G. Wang, J. Xu, and X. Du, "Survey on Data Management in Blockchain Systems," *Journal of Software.*, vol. 09, no. 31, 2020, pp. 2903-2925.
- [101] B. Behera, "Sorting an Array Using the Topological Sort of a Corresponding Comparison Graph," *Theor. Comput. Sci.*, 2020, pp. 76-97.
- [102] Thegraph, "The Graph Network: Subgraph Migration Is in Full Swin," 2022, Technical Report. <https://thegraph.com/blog/graph-network-migration>
- [103] X. Tong, H. Tang, N. Jiang, W. Fan, Y. Gao, S. Deng, Z. Zhang, C. Jin, Y. Yang, and G. Qin, "SQL-Middleware: Enabling the Blockchain with SQL," *DASFAA.*, vol. 3, pp. 622-626, 2021.
- [104] J. Han, H. Kim, H. Eom, J. Coignard, K. Wu, and Y. Son, "Enabling SQL-Query Processing for Ethereum-based Blockchain Systems," *WIMS.*, 2019, pp. 1-7.
- [105] V. Buterin, "Chain interoperability," 2019, Technical Report. <https://www.r3.com/download/chain-interoperability>



HANYU MAO received the bachelor's degrees in computer science from Northeastern University, China, where he is currently pursuing the Ph.D. degree. His research interests include distributed database and blockchain.



TIEZHENG NIE received the Ph.D. degree in computer science from Northeastern University, Shenyang, China, in 2009. He is currently an Associate Professor with the School of Computer Science and Engineering, Northeastern University. He is a member of CCF. His main research interests include blockchain and big data management.



SUN HAO received the bachelor's and master's degrees in computer science from Northeastern University, China. Her research interests include distributed database and blockchain.



DERONG SHEN (Member, IEEE) received the Ph.D. degree in computer science from Northeastern University, Shenyang, China, in 2004. She is currently a Professor and a Ph.D. Supervisor with the School of Computer Science and Engineering, Northeastern University. She is a Senior Member of CCF. Her research interests include blockchain and data integration.



GE YU (Senior Member, IEEE) received the Ph.D. degree in computer science from Kyushu University, Japan, in 1996. He is currently a Professor with Northeastern University, China. He has published more than 200 papers in refereed journals and conferences. His current research interests include distributed and parallel systems, cloud computing and big data management, and blockchain techniques and systems. He is a fellow of CCF and a member of ACM.